

DD: SUB-WF-DOWNGRADE-01 — Subscription Downgrade

Developer implementation guide:

DD_SUB-WF-DOWNGRADE-01-Subscription_Downgrade-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract

DD_SUB-WF-DOWNGRADE-01-Subscription_Downgrade-v1.0.md.

Verdict: ■ New | **Wave:** 1 | **Group I:** Subscription Management | **Version:** v1.0 Operation that transitions an ACTIVE subscription's package_ref from a higher-priced Package to a lower-priced Package on the same HomePass. Owns: the 2-trigger taxonomy (CUSTOMER_REQUESTED_DOWNGRADE / ADMIN_DOWNGRADE), the target-package validation rules (currency match, price ≤ source, ACTIVE status, HomePass service-class compatibility), the equipment-unbind logic (target Package may drop service classes — e.g., triple→double-play returns STB), the cycleAnchorPolicy + effectiveTiming taxonomies (mirrors UPGRADE-01), the per-operator config catalog, the trigger REST endpoint + cancel endpoint, the rich domain Kafka events. Composes the framework primitives from **DD_SUB-WF-FRAMEWORK-01** + **DD_SUB-WF-FRAMEWORK-01-WORKERS**. **Effective timing — customer choice per request.** Three modes: IMMEDIATE (downgrade takes effect at saga commit), END_OF_CURRENT_CYCLE (downgrade fires at the cycle close timestamp), SCHEDULED_AT (downgrade fires at an arbitrary future timestamp within subscription_downgrade_config.max_future_scheduled_days). **Default is** END_OF_CURRENT_CYCLE — the typical customer-friendly behavior: customer paid for the cycle at higher tier, gets the value through cycle end, downgrade applies at next cycle. The BPMN parks on a Camunda intermediate timer until the chosen moment, then runs validate-bill-fulfill commit. Scheduled-future downgrades are cancellable via DELETE. **Distinct from UPGRADE-01:** price comparison reversed (target ≤ source). BIL-01 returns a *credit* (signed negative) for IMMEDIATE / SCHEDULED_AT mid-cycle commits with PRESERVE policy, or zero delta for END_OF_CURRENT_CYCLE + PRESERVE. Equipment-unbind is **enabled by default** for downgrades (obsolete equipment returned to inventory), whereas UPGRADE-01 retains existing equipment. **Companion DDs** (per-operator BPMN satellites): - **DD_SUB-WF-DOWNGRADE-01-FLOW-WIK** — KE WIK reference BPMN (shipped v1.0) - **DD_SUB-WF-DOWNGRADE-01-FLOW-WUG / -WTZ / -YASN** — pending operator deep dives

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/04_subscription_workflows/DD_SUB-WF-DOWNGRADE-01-Subscription_Downgrade-v1.0.md
Version	v1.0
DD type	module contract

Item	Value
BPMN/process keys	Not explicitly defined
Drools packages	rules.subscription.downgrade.<operator>

4. Runtime Contracts To Implement

4.1 APIs

- DELETE /api/subscriptions/{subscription_id}/pending-downgrade/{operation_id}
- DELETE /pending-downgrade/{operationId}
- POST /api/subscriptions/{subscription_id}/downgrade

4.2 Tables

- subscription_downgrade_config

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-emit-intent
- drools-decide
- external-cancel-requested
- ful-call-modify
- kafka-emit
- notification-send
- osr-equipment-unbind
- sub-lm-commit-state
- sub-lm-compute-cycle-window
- sub-lm-put-master-fields
- sub-lm-put-pending-status
- subscription-operation-finalize

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- SubscriptionDowngradeCancelled
- SubscriptionDowngradeRejected
- SubscriptionDowngradeScheduled
- SubscriptionDowngraded

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

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Effective timing — customer choice per request. Three modes: IMMEDIATE (downgrade takes effect at saga commit), END_OF_CURRENT_CYCLE (downgrade fires at the cycle close timestamp), SCHEDULED_AT (downgrade fires at an arbitrary future timestamp within subscription_downgrade_config.max_future_scheduled_days). **Default is** END_OF_CURRENT_CYCLE — the typical customer-friendly behavior: customer paid for the cycle at higher tier, gets the value through cycle end, downgrade applies at next cycle. The BPMN parks on a Camunda intermediate timer until the chosen moment, then runs validate-bill-fulfill commit. Scheduled-future downgrades are cancellable via DELETE.

Distinct from UPGRADE-01: price comparison reversed (target ≤ source). BIL-01 returns a *credit* (signed negative) for IMMEDIATE / SCHEDULED_AT mid-cycle commits with PRESERVE policy, or zero delta for END_OF_CURRENT_CYCLE + PRESERVE. Equipment-unbind is **enabled by default** for downgrades (obsolete equipment returned to inventory), whereas UPGRADE-01 retains existing equipment.

Companion DDs (per-operator BPMN satellites):

- **DD_SUB-WF-DOWNGRADE-01-FLOW-WIK** — KE WIK reference BPMN (shipped v1.0)
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Glossary

Term	Meaning
Downgrade	Change of the subscription's Package reference to a lower-priced Package on the same HomePass. Distinct from MIGRATION (different HomePass) and UPGRADE (higher price).
Source Package	The current package_ref + package_version_id snapshot at downgrade time. Pinned to the row for audit; also written to previous_package_ref on the subscription.
Target Package	The Package the subscription is downgrading to. Must be ACTIVE in SIP-01, same currency as the subscription, price ≤ source Package price, and every service family it requires must be present in the HomePass's service_management_endpoints (RLM-CFG-01) with a non-null port.
Credit delta	The amount BIL-01 computes (negative — credit to customer) for the remaining days of the current cycle at the price difference between source and target Packages. For prepaid: applied as wallet top-up. For postpaid: invoice credit line. Owned by BIL-01 + BIL-CFG-01.
Equipment-drop delta	The set of service classes in the source Package that are NOT in the target Package. Equipment bound to those service classes (e.g., STB for IPTV) is candidate for unbind + return per unbindObsoleteEquipment.
cycleAnchorPolicy	Customer/admin choice captured per downgrade request. Two values: PRESERVE (keep existing cycle anchor; for IMMEDIATE/mid-cycle commits BIL-01 issues credit delta for cycle-days-remaining) or RESET_TO_DOWNGRADE_DATE (cycle anchor moves to commit date; BIL-01 issues old-cycle credit + fresh first-cycle at target price). Operator config may default. Applied at actual commit time.
effectiveTiming	Customer choice per request: IMMEDIATE (commit at saga start), END_OF_CURRENT_CYCLE (commit at the current cycle's cycle_end; the default for downgrades), SCHEDULED_AT (commit at a customer-supplied future timestamp). For non-IMMEDIATE, BPMN parks on a Camunda intermediate timer. The subscription remains ACTIVE on the source Package during the wait; charges continue at source price.
Pending-future downgrade	A scheduled downgrade in its wait state. Represented by the BPMN process instance parked on the timer + the subscription_operation row with current_state = SCHEDULED. Cancellable via DELETE.

1. What this is

The downgrade operation. SUB-WF-DOWNGRADE-01 is one of the 10 operation kinds in the framework. Triggered by: a customer requesting a downgrade (CC, portal, retail) or an admin downgrading on the customer's behalf (e.g., as part of a retention save).

Outcomes (at actual commit time, which may be NOW or a future moment per effectiveTiming): subscription package_ref + package_version_id updated to target Package, previous_package_ref snapshot retained, credit applied per cycleAnchorPolicy (or zero delta for END_OF_CURRENT_CYCLE + PRESERVE), network reconfigured for new Service profile (FUL-03 MODIFY_SERVICE), equipment-unbind step run if equipment-drop delta exists AND unbindObsoleteEquipment = true, cycle anchor preserved or reset per the operation's chosen policy, customer notified.

For non-IMMEDIATE effectiveTiming, the subscription remains ACTIVE on the source Package during the wait. A second pending-future downgrade against the same subscription is rejected; any state-affecting operation that fires during the wait cancels the pending downgrade per R-EF-5.

What this DD owns

1. **subscription_downgrade_config table** — per-operator config: customer-self-service enablement, default cycle-anchor policy, default effective timing, max-future-scheduled-days, default equipment-unbind behavior.
2. **Trigger taxonomy** — 2 triggers (CUSTOMER_REQUESTED_DOWNGRADE / ADMIN_DOWNGRADE) with role-gating.
3. **Target-package validation rules** — currency match, status ACTIVE, price ≤ source, HomePass compatibility, equipment-drop delta computation.
4. **cycleAnchorPolicy taxonomy** — 2 values (PRESERVE / RESET_TO_DOWNGRADE_DATE) carried per request; defaulted by operator config.
5. **effectiveTiming taxonomy** — 3 values (IMMEDIATE / END_OF_CURRENT_CYCLE / SCHEDULED_AT); default = END_OF_CURRENT_CYCLE; for non-IMMEDIATE the BPMN parks on a Camunda intermediate timer; cancellable.
6. **Trigger REST endpoints** — POST /api/subscriptions/{subscription_id}/downgrade to start a downgrade; DELETE /api/subscriptions/{subscription_id}/pending-downgrade/{operation_id} to cancel a pending-future downgrade.
7. **Domain Kafka events** — SubscriptionDowngradeScheduled (non-IMMEDIATE acceptance), SubscriptionDowngraded (commit), SubscriptionDowngradeRejected (failure), SubscriptionDowngradeCancelled (cancellation of pending-future).
8. **Downgrade workflow rules** — 37 rules in §4.

What this DD does NOT own

- **BPMN files** — owned by per-operator satellites.
- **Worker implementations** — catalogued in DD_SUB-WF-FRAMEWORK-01-WORKERS.
- **Subscription state machine** — owned by SUB-LM-01.
- **Package catalog** — owned by SIP-01.
- **Service catalog + service-class definitions + equipment-requirement declarations** — owned by PLM-CFG-01.
- **HomePass row mutations** — owned by RLM-CFG-01. This workflow reads service_management_endpoints, status_code, and the status-code catalog; it does not mutate any HomePass column. If the target Package requires a service family the HomePass doesn't physically support, the downgrade is blocked at validation — caller must use SUB-WF-MIGRATION-01.
- **Credit delta math + refund mechanism (wallet top-up vs invoice credit) + BillableEvent definitions** — owned by BIL-CFG-01 + BIL-01.
- **Network reconfiguration + per-service-class provisioner fan-out** — owned by FUL-03 under MODIFY_SERVICE. This workflow sends FUL-03 the source + target Package snapshots; FUL-03 computes the service-class diff and dispatches the appropriate provisioner per service class (modifyService for unchanged, deactivate for dropped, activate for any newly-required — typically none on downgrade).
- **Equipment binding mechanics + return-to-inventory** — owned by OSR-INSTANCE-01 + OSR-RMA-01 (return material authorization for physically-recovered equipment).
- **Restrictions during downgrade** — preserved as-is on the subscription's active_restrictions[] array; this DD does NOT clear or re-apply restrictions.
- **Customer notification template + channel** — owned by NOT-01.

Regional flexibility

Per-operator levers:

Lever	Examples for DOWNGRADE
BPMN file	KE: standard BPMN with conditional equipment-unbind step + credit-distribution step. Other operators may add a supervisor-approval user task for high-value retention saves.
Drools package rules.subscription.downgrade.<operator>	KE: target-package validation, equipment-drop-delta computation, self-service eligibility.
subscription_downgrade_config row	KE: customer self-service enabled; default END_OF_CURRENT_CYCLE + PRESERVE + unbind enabled.
FOUNDATION_AUTH candidate groups	Operator-specific resolution.

2. Tables overview

This DD owns one table: subscription_downgrade_config (per-operator config).

Cross-operation audit anchor: subscription_operation (framework-owned).

Subscription master fields updated (owned by SUB-LM-01): package_ref, package_version_id, previous_package_ref, previous_package_version_id, last_status_changed_at.
current_transition_type = DOWNGRADE during PENDING_DOWNGRADE.

3. Schemas

3.1 subscription_downgrade_config

One row per operator.

```
CREATE TABLE subscription_downgrade_config (  
  operator_code TEXT PRIMARY KEY,  
  customer_self_service_enabled BOOLEAN NOT NULL DEFAULT FALSE,  
  default_cycle_anchor_policy TEXT NOT NULL DEFAULT 'PRESERVE',  
  default_effective_timing TEXT NOT NULL DEFAULT 'END_OF_CURRENT_CYCLE',  
  max_future_scheduled_days INTEGER NOT NULL DEFAULT 90,  
  default_unbind_obsolete_equipment BOOLEAN NOT NULL DEFAULT TRUE,  
  customer_notification_enabled BOOLEAN NOT NULL DEFAULT TRUE,  
  updated_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),  
  updated_by TEXT NOT NULL,  
  CHECK (default_cycle_anchor_policy IN ('PRESERVE', 'RESET_TO_DOWNGRADE_DATE')),  
  CHECK (default_effective_timing IN ('IMMEDIATE', 'END_OF_CURRENT_CYCLE', 'SCHEDULED_AT')),  
  CHECK (max_future_scheduled_days > 0)  
);
```

Column	Notes
customer_self_service_enabled	When true, the CUSTOMER_REQUESTED_DOWNGRADE trigger may be initiated from the customer portal (CUSTOMER_SELF_SERVICE role). When false, only CALL_CENTER and admin roles may initiate it.
default_cycle_anchor_policy	Default cycleAnchorPolicy applied when the request does not supply one. PRESERVE keeps the existing cycle anchor. RESET_TO_DOWNGRADE_DATE moves cycle anchor to commit date.
default_effective_timing	Default effectiveTiming applied when the request does not supply one. END_OF_CURRENT_CYCLE is the standard downgrade default (customer paid for the cycle, gets full value through cycle end).
max_future_scheduled_days	Maximum window (days from request time) for SCHEDULED_AT.scheduledAt. Requests beyond this return 400 SCHEDULED_AT_BEYOND_MAX_WINDOW.
default_unbind_obsolete_equipment	When true (default for downgrades), the workflow unbinds equipment that was bound to service classes the target Package no longer requires (e.g., return STB on triple→double-play). The request may override. Equipment physical recovery is OSR-RMA-01's responsibility.
customer_notification_enabled	When true, the workflow calls notification-send at acceptance (for scheduled) AND at commit. NOT-01 owns template, channel, language, fallback.

Sample data:

operator_code	customer_self_service_enabled	default_cycle_anchor_policy	default_effective_timing	max_future_scheduled_days	default_unbind_obsolete_equipment	customer_notification_enabled
WIK	true	PRESERVE	END_OF_CURRENT_CYCLE	90	true	true
WUG	false	PRESERVE	END_OF_CURRENT_CYCLE	90	true	true
WTZ	false	PRESERVE	END_OF_CURRENT_CYCLE	90	true	true
YASSN	false	PRESERVE	END_OF_CURRENT_CYCLE	90	true	true

4. Rules

4.1 State precondition + transition

Rule	Description
R-SUB-WF-DOWNGRADE-01-S-1	Downgrade requires the subscription's <code>status_code = ACTIVE</code> at request time AND at commit time. Other statuses return 409 <code>INVALID_STATE_FOR_DOWNGRADE</code> . For scheduled-future downgrades, the check at commit time may fail if the subscription was paused or terminated during the wait — handled by R-EF-5.
R-SUB-WF-DOWNGRADE-01-S-2	The transition at commit is <code>ACTIVE → PENDING_DOWNGRADE → ACTIVE</code> . <code>PENDING_DOWNGRADE</code> is the framework-internal transient state during the commit sequence. For scheduled-future downgrades, the subscription stays <code>ACTIVE</code> during the wait — <code>PENDING_DOWNGRADE</code> is entered only when the timer fires.
R-SUB-WF-DOWNGRADE-01-S-3	Per framework R-FW-1, downgrade <code>COMMIT</code> is mutually exclusive with other state-affecting operations on the same subscription. A second downgrade <code>COMMIT</code> cannot fire while another commit is in progress. A scheduled-future downgrade in its wait state DOES allow other operations to fire (e.g., customer-requested <code>PAUSE</code>) — see R-EF-5 for cancellation semantics.
R-SUB-WF-DOWNGRADE-01-S-4	The subscription's <code>current_transition_type</code> is set to <code>DOWNGRADE</code> during <code>PENDING_DOWNGRADE</code> . <code>current_transition_reason_code</code> is set to the supplied <code>downgradeTrigger</code> . NOT set during scheduled-future wait.
R-SUB-WF-DOWNGRADE-01-S-5	Restrictions on the subscription (<code>active_restrictions[]</code>) are NEITHER cleared NOR validated by this workflow. A customer with a restriction MAY downgrade — the restriction persists post-downgrade.
R-SUB-WF-DOWNGRADE-01-S-6	At most one pending-future downgrade per subscription. A second non- <code>IMMEDIATE</code> downgrade request returns 409 <code>PENDING_DOWNGRADE_ALREADY_SCHEDULED</code> with the existing operation id. An <code>IMMEDIATE</code> downgrade request while a pending-future downgrade exists returns 409 (caller must cancel the pending one first or wait).

4.2 Trigger taxonomy + role gating

Rule	Description
R-SUB-WF-DOWNGRADE-01-RT-1	2 triggers: <code>CUSTOMER_REQUESTED_DOWNGRADE / ADMIN_DOWNGRADE</code> . Carried in the request body as <code>downgradeTrigger</code> . Unknown values return 400 <code>UNKNOWN_DOWNGRADE_TRIGGER</code> .
R-SUB-WF-DOWNGRADE-01-RT-2	Role-trigger gating: <code>CUSTOMER_REQUESTED_DOWNGRADE</code> requires <code>CUSTOMER_SELF_SERVICE</code> OR <code>CALL_CENTER</code> OR <code>SUBSCRIPTION_OPERATOR</code> ; <code>ADMIN_DOWNGRADE</code> requires <code>SUBSCRIPTION_ADMIN</code> OR <code>SUBSCRIPTION_OPERATOR</code> . Mismatched role returns 403 <code>ROLE_NOT_AUTHORIZED_FOR_TRIGGER</code> .
R-SUB-WF-DOWNGRADE-01-RT-3	<code>CUSTOMER_SELF_SERVICE</code> role allowed only when <code>subscription_downgrade_config.customer_self_service_enabled = true</code> . Else 403.

4.3 Target-package validation

Rule	Description
R-SUB-WF-DOWNGRADE-01-TP-1	<code>targetPackageRef</code> must reference a package row with <code>status = ACTIVE</code> in SIP-01. <code>DRAFT</code> , <code>INACTIVE</code> , or <code>END_OF_LIFE</code> Packages return 400 <code>TARGET_PACKAGE_NOT_ACTIVE</code> .
R-SUB-WF-DOWNGRADE-01-TP-2	The target Package's current active <code>package_version</code> row supplies the price + currency snapshot. Currency MUST match the subscription's currency. Cross-currency downgrades return 400 <code>CURRENCY_MISMATCH</code> .
R-SUB-WF-DOWNGRADE-01-TP-3	Target Package's current active version price MUST be $\leq$ source Package's pinned version price. Higher-priced target returns 400 <code>NOT_A_DOWNGRADE</code> with guidance to use <code>SUB-WF-UPGRADE-01</code> .
R-SUB-WF-DOWNGRADE-01-TP-4	Target Package MUST be different from source Package (<code>targetPackageRef</code> $\neq$ <code>subscription.package_ref</code>). Same-package returns 400 <code>SAME_PACKAGE</code> .

Rule	Description
R-SUB-WF-DOWNGRADE-01-TP-5	The current HomePass MUST be physically capable of carrying every service family required by the target Package. Specifically: for each Service in the target Package's <code>package_composition</code> , the Service's <code>serviceManagementKey</code> (e.g., <code>data</code> , <code>voice</code> , <code>iptv_multicast</code>) MUST be present in <code>homepass.service_management_endpoints</code> keys. If a required key is absent, return 400 <code>PACKAGE_NOT_AVAILABLE_AT_HOMEPASS</code> with guidance to use <code>SUB-WF-MIGRATION-01</code> . Note: for typical downgrades the target Package's required keys are a subset of the source's, so this check is rarely the blocker; included for symmetry and to catch edge cases (e.g., target Package adds a service family not in source).
R-SUB-WF-DOWNGRADE-01-TP-6	For each required <code>serviceManagementKey</code> , the resolved endpoint's port MUST NOT be null. Null port returns 400 <code>HOMEPASS_PORT_NOT_PROVISIONED</code> .
R-SUB-WF-DOWNGRADE-01-TP-7	The HomePass <code>status_code</code> MUST be in a status with <code>is_active = true</code> per the operator's <code>homepass_status_code</code> catalog. HomePass in retired, suspended, or not-yet-serviceable status returns 409 <code>HOMEPASS_NOT_ACTIVE</code> .
R-SUB-WF-DOWNGRADE-01-TP-8	Target Package's <code>billing-frequency-days</code> SHOULD match the source Package's. Mismatch is not rejected but produces a warning event field <code>billingFrequencyChanged=true</code> ; BIL-01 handles cycle implications per its policy.

4.4 Equipment-drop delta

Rule	Description
R-SUB-WF-DOWNGRADE-01-EQ-1	The workflow computes the set of service classes in the source Package that are NOT in the target Package. For each obsolete service class, the workflow inspects existing equipment bindings (<code>OSR-INSTANCE-01</code>) to identify equipment that fulfills only that obsolete service class. The list is <code>obsoleteEquipmentList</code> .
R-SUB-WF-DOWNGRADE-01-EQ-2	When <code>unbindObsoleteEquipment = true</code> (the request value, defaulting from operator config), the BPMN runs the equipment-unbind step for each entry in <code>obsoleteEquipmentList</code> . <code>OSR-INSTANCE-01</code> marks the equipment as returned-to-inventory; physical recovery (RMA, pickup WO) is <code>OSR-RMA-01</code> 's responsibility.
R-SUB-WF-DOWNGRADE-01-EQ-3	When <code>unbindObsoleteEquipment = false</code> , equipment stays bound to the subscription post-downgrade. The Service Profile sent to <code>FUL-03</code> drops the obsolete service class, so the equipment is provisioning-orphaned — it's vestigial. Operator policy decides; some operators leave the STB with the customer in case they upgrade back later.
R-SUB-WF-DOWNGRADE-01-EQ-4	Equipment that fulfills BOTH an obsolete service class AND a retained service class is NOT unbound. Example: a multi-service ONT fulfilling both data and voice — when downgrading to data-only, the ONT stays bound (still fulfills data).

4.5 Billing + fulfillment

Rule	Description
R-SUB-WF-DOWNGRADE-01-B-1	Workflow calls <code>bil01-emit-intent</code> with <code>intentKind = DOWNGRADE_INTENT</code> . Context: <code>{sourcePackageRef, sourcePackageVersionId, sourcePackagePrice, targetPackageRef, targetPackageVersionId, targetPackagePrice, billingFrequencyChanged, downgradeTrigger, cycleAnchorPolicy, effectiveTiming}</code> . BIL-01 computes the appropriate credit per <code>cycleAnchorPolicy</code> (see <code>R-CA-3</code>), may apply <code>BillableEvents</code> per <code>BIL-CFG-01</code> .
R-SUB-WF-DOWNGRADE-01-B-2	Pay-first gate does NOT apply (downgrade is value-decreasing — never produces a positive customer charge). Credit distribution per BIL-01's <code>billing-mode</code> policy: <code>prepaid</code> → wallet top-up; <code>postpaid</code> → invoice credit line. BIL-01 owns the mechanism.
R-SUB-WF-DOWNGRADE-01-B-3	<code>FUL-03</code> called with <code>intent=MODIFY_SERVICE</code> , passing source + target Package snapshots. <code>FUL-03</code> owns the per-service-class fan-out: (a) Service classes in both source and target → <code>provisioner modifyService</code> (or no-op if Service profile unchanged); (b) service classes in source but NOT target → <code>provisioner deactivate</code> , dispatched via the HomePass's <code>service_management_endpoints[serviceManagementKey]</code> ; (c) service classes in target but NOT source → <code>provisioner activate</code> (rare on downgrade; possible if customer is switching service mix). <code>FUL-03</code> handles per-customer network-side teardown (IGMP-snooping disable, multicast group leave, AAA channel-package removal) under the relevant <code>provisioner deactivate</code> for dropped service classes.

Rule	Description
R-SUB-WF-DOWNGRADE-01-B-4	HomePass row is NOT mutated by this workflow. The HomePass's <code>service_management_endpoints</code> and <code>status_code</code> are RLM-CFG-01-owned read-only inputs. What changes during a service-class-drop downgrade is *per-customer network-side teardown* (FUL-03's responsibility) and *equipment unbinding* on the subscription (this DD's §4.4).
R-SUB-WF-DOWNGRADE-01-B-5	On commit, master fields updated: <code>package_ref</code> = <code>targetPackageRef</code> , <code>package_version_id</code> = <code>targetPackageVersionId</code> , <code>previous_package_ref</code> = <code>sourcePackageRef</code> , <code>previous_package_version_id</code> = <code>sourcePackageVersionId</code> , <code>last_status_changed_at</code> = NOW. If <code>cycleAnchorPolicy</code> = RESET_TO_DOWNGRADE_DATE: <code>cycle_anchor_day</code> and <code>cycle_anchor_at</code> master fields are also updated to NOW.

4.6 Cycle anchor policy

Rule	Description
R-SUB-WF-DOWNGRADE-01-CA-1	The request supplies <code>cycleAnchorPolicy</code> ∈ {PRESERVE, RESET_TO_DOWNGRADE_DATE}. If absent, the workflow defaults to <code>subscription_downgrade_config.default_cycle_anchor_policy</code> for the operator. Unknown value returns 400 UNKNOWN_CYCLE_ANCHOR_POLICY.
R-SUB-WF-DOWNGRADE-01-CA-2	<code>cycleAnchorPolicy</code> is captured per downgrade — NOT a subscription-level invariant. The resolved policy is persisted to the Kafka event payload + the framework's <code>subscription_operation</code> audit row.
R-SUB-WF-DOWNGRADE-01-CA-3	BIL-01 interprets <code>cycleAnchorPolicy</code> in the DOWNGRADE_INTENT context. PRESERVE → BIL-01 returns a prorated credit line item (DOWNGRADE_PRORATE_CREDIT BillableEvent — negative amount) for <code>cycle-days-remaining</code> × <code>price-difference</code> . RESET_TO_DOWNGRADE_DATE → BIL-01 returns an old-cycle credit line (DOWNGRADE_OLD_CYCLE_CREDIT) for the unused portion of the source-priced cycle + a fresh first-cycle line (DOWNGRADE_FIRST_CYCLE — positive at target price) for the new cycle starting at commit time. Net is still ≤ 0 (downgrade never charges the customer more).
R-SUB-WF-DOWNGRADE-01-CA-4	When RESET_TO_DOWNGRADE_DATE, the new cycle starts at commit time and runs for <code>targetPackage.billing_frequency_days</code> . Subsequent cycles roll forward from the new anchor. Cycle-anchor mutation is committed via the master-fields step alongside <code>package_ref</code> .

4.7 Effective timing

Rule	Description
R-SUB-WF-DOWNGRADE-01-EF-1	The request supplies <code>effectiveTiming</code> ∈ {IMMEDIATE, END_OF_CURRENT_CYCLE, SCHEDULED_AT}. If absent, the workflow defaults to <code>subscription_downgrade_config.default_effective_timing</code> (KE v1.0: END_OF_CURRENT_CYCLE). Unknown value returns 400 UNKNOWN_EFFECTIVE_TIMING.
R-SUB-WF-DOWNGRADE-01-EF-2	When <code>effectiveTiming</code> = SCHEDULED_AT, the request MUST also supply <code>scheduledAt</code> (ISO 8601 timestamp). Missing → 400 SCHEDULED_AT_MISSING. <code>scheduledAt</code> MUST be strictly in the future AND within <code>max_future_scheduled_days</code> from now. Past or beyond-window → 400 SCHEDULED_AT_OUT_OF_BOUNDS.
R-SUB-WF-DOWNGRADE-01-EF-3	Combination <code>effectiveTiming</code> = END_OF_CURRENT_CYCLE AND <code>cycleAnchorPolicy</code> = RESET_TO_DOWNGRADE_DATE is redundant. Returns 400 REDUNDANT_POLICY_COMBINATION with guidance to use <code>cycleAnchorPolicy</code> = PRESERVE for end-of-cycle downgrades.
R-SUB-WF-DOWNGRADE-01-EF-4	For non-IMMEDIATE timings: the BPMN parks on a Camunda intermediate timer. The <code>subscription_operation.current_state</code> is set to SCHEDULED. The trigger endpoint returns 202 with <code>scheduledCommitAt</code> populated. <code>SubscriptionDowngradeScheduled</code> is emitted on Kafka. At the timer fire, the BPMN re-validates and proceeds with the commit sequence.
R-SUB-WF-DOWNGRADE-01-EF-5	Cancellation paths during the wait: (a) explicit DELETE /pending-downgrade/{operationId} from BO admin or customer — stops the timer, emits <code>SubscriptionDowngradeCancelled</code> , closes the operation as CANCELLED; (b) any state-affecting operation on the same subscription (PAUSE, SUSPEND-NP, TERMINATE, UPGRADE, another DOWNGRADE) — the framework's external-cancel-requested mechanism stops the timer and emits <code>SubscriptionDowngradeCancelled</code> with <code>cancellationReasonCode</code> = SUPERSEDED_BY_<otherOperationKind>.

Rule	Description
R-SUB-WF-DOWNGRADE-01-EF-6	At commit time (after the wait), the workflow re-validates all preconditions (S-1, TP-1 through TP-7, equipment-drop delta still consistent). If any check fails at commit, the workflow rejects with <code>SubscriptionDowngradeRejected</code> carrying <code>rejectionStage = COMMIT_REVALIDATION</code> . Example: target Package transitioned to <code>END_OF_LIFE</code> between schedule and commit → reject.
R-SUB-WF-DOWNGRADE-01-EF-7	The <code>cycleAnchorPolicy</code> recorded at request time is honored at commit time. For <code>END_OF_CURRENT_CYCLE + PRESERVE</code> , the commit naturally lands on the cycle boundary; BIL-01 returns zero delta (the current cycle finishes on source price, next cycle starts on target price); no anchor mutation needed. For <code>SCHEDULED_AT</code> mid-cycle commits, BIL-01 behaves identically to <code>IMMEDIATE</code> at the commit moment (credit delta if <code>PRESERVE</code> ; old-cycle credit + fresh cycle if <code>RESET_TO_DOWNGRADE_DATE</code> — anchor moves to the scheduled date).
R-SUB-WF-DOWNGRADE-01-EF-8	Credit settlement is performed at commit time (not request time) — for non- <code>IMMEDIATE</code> downgrades, the customer's wallet / invoice is only touched at commit. This matches the principle that scheduled downgrades shouldn't pre-issue credits (the customer keeps using the higher-tier service through the wait).

5. REST API

5.1 Trigger downgrade — IMMEDIATE example

```
POST /api/subscriptions/{subscription_id}/downgrade HTTP/1.1
Authorization: Bearer <jwt>
Content-Type: application/json
Idempotency-Key: downgrade-sub_01HZ_KAMAU-pkg_INET_100-20260520

{
  "operatorCode":      "WIK",
  "downgradeTrigger":  "CUSTOMER_REQUESTED_DOWNGRADE",
  "targetPackageRef":  "pkg_01HX_INET_100M_VOIP_RES",
  "cycleAnchorPolicy": "PRESERVE",
  "effectiveTiming":   "IMMEDIATE",
  "scheduledAt":       null,
  "unbindObsoleteEquipment": true,
  "notes":             "Customer downgrading after promotional period ended",
  "initiatingActorUserId": "cc-agent-wik-007",
  "initiatingActorRole":  "CALL_CENTER",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-05-20-5511"
}

→ 202 Accepted
{
  "operationId":      "subop_01HZ_KAMAU_DOWNGRADE",
  "operatorCode":     "WIK",
  "subscriptionId":   "sub_01HZ_KAMAU_FIBER",
  "operationKind":    "DOWNGRADE",
  "downgradeTrigger": "CUSTOMER_REQUESTED_DOWNGRADE",
  "resolvedCycleAnchorPolicy": "PRESERVE",
  "resolvedEffectiveTiming":  "IMMEDIATE",
  "scheduledCommitAt": null,
  "bpmnProcessKey":   "sub-downgrade-wik",
  "bpmnProcessInstanceId": "pi_01HZ_KAMAU_DOWNGRADE",
  "currentState":     "VALIDATING",
  "startedAt":        "2026-05-20T14:00:00+03:00"
}
```

5.2 Trigger downgrade — END_OF_CURRENT_CYCLE (the typical default)

```
POST /api/subscriptions/{subscription_id}/downgrade HTTP/1.1

{
  "operatorCode":      "WIK",
  "downgradeTrigger":  "CUSTOMER_REQUESTED_DOWNGRADE",
  "targetPackageRef":  "pkg_01HX_INET_100M_VOIP_RES",
  "unbindObsoleteEquipment": true,
  ...
}

→ 202 Accepted
{
  "operationId":      "subop_01HZ_KAMAU_DOWNGRADE_EOC",
  "resolvedCycleAnchorPolicy": "PRESERVE",
  "resolvedEffectiveTiming":  "END_OF_CURRENT_CYCLE",
}
```

```

    "scheduledCommitAt": "2026-05-31T23:59:59+03:00",
    "currentState": "SCHEDULED"
}

```

(Note: the request omitted both `cycleAnchorPolicy` and `effectiveTiming`; the workflow resolved both from operator-config defaults — PRESERVE + END_OF_CURRENT_CYCLE for KE.)

5.3 Errors

Not a downgrade (target more expensive):

```

→ 400 Bad Request
{
  "code": "NOT_A_DOWNGRADE",
  "sourcePackagePrice": 2500.00,
  "targetPackagePrice": 4500.00,
  "currency": "KES",
  "guidance": "Target Package is more expensive. Use SUB-WF-UPGRADE-01."
}

```

Redundant policy combination:

```

→ 400 Bad Request
{
  "code": "REDUNDANT_POLICY_COMBINATION",
  "effectiveTiming": "END_OF_CURRENT_CYCLE",
  "cycleAnchorPolicy": "RESET_TO_DOWNGRADE_DATE",
  "guidance": "END_OF_CURRENT_CYCLE commit naturally aligns the new anchor with cycle close. Use cycleAnchorPolicy"
}

```

Already-scheduled downgrade:

```

→ 409 Conflict
{
  "code": "PENDING_DOWNGRADE_ALREADY_SCHEDULED",
  "existingOperationId": "subop_01HZ_KAMAU_DOWNGRADE_EOC",
  "existingScheduledCommitAt": "2026-05-31T23:59:59+03:00",
  "guidance": "Cancel the existing scheduled downgrade before submitting a new one."
}

```

5.4 Cancel pending-future downgrade

DELETE /api/subscriptions/{subscription_id}/pending-downgrade/{operation_id} HTTP/1.1

```

{
  "operatorCode": "WIK",
  "cancellationReasonCode": "CUSTOMER_CHANGED_MIND",
  "notes": "Customer called to cancel – wants to keep current plan",
  "initiatingActorUserId": "cc-agent-wik-007",
  "initiatingActorRole": "CALL_CENTER"
}

→ 200 OK
{
  "operationId": "subop_01HZ_KAMAU_DOWNGRADE_EOC",
  "cancellationReasonCode": "CUSTOMER_CHANGED_MIND",
  "cancelledAt": "2026-05-22T11:30:00+03:00",
  "subscriptionStaysOn": "pkg_01HX_INET_250M_VOIP_RES",
  "finalState": "CANCELLED"
}

```

6. Kafka events emitted

Event	Topic	Fires on	Consumers
SubscriptionDowngradeScheduled	sophix.subscription.subscription.downgradescheduled	Acceptance of non-IMMEDIATE timing	NOT-01 (customer notification "your downgrade is scheduled for ..."), CVM (intent funnel), EM-04
SubscriptionDowngraded	sophix.subscription.subscription.downgraded	Successful commit (IMMEDIATE OR after timer fires)	NOT-01 (customer notification per operator config), BIL-02 (cycle accrual at new price), CVM (downgrade/churn analytics), EM-04 reporting, SIP-01 (package-version-pinning audit), OSR-RMA-01 (kicks off equipment-return WO for unbound equipment)

Event	Topic	Fires on	Consumers
SubscriptionDowngradeCancelled	sophix.subscription.subscription.downgradecancelled	Pending-future downgrade cancelled	NOT-01, CVM, EM-04
SubscriptionDowngradeRejected	sophix.subscription.subscription.downgraderejected	Workflow failure (validation, billing, fulfillment, commit-time revalidation)	NOT-01, ICN-01 ops alert, EM-04

Sample SubscriptionDowngradeScheduled:

```
{
  "eventType": "SubscriptionDowngradeScheduled",
  "aggregateId": "sub_01HZ_KAMAU_FIBER",
  "timestamp": "2026-05-20T14:00:01+03:00",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "customerId": "cust_01HZ_KAMAU",
    "operationId": "subop_01HZ_KAMAU_DOWNGRADE_EOC",
    "downgradeTrigger": "CUSTOMER_REQUESTED_DOWNGRADE",
    "sourcePackageRef": "pkg_01HX_INET_250M_VOIP_RES",
    "targetPackageRef": "pkg_01HX_INET_100M_VOIP_RES",
    "resolvedCycleAnchorPolicy": "PRESERVE",
    "resolvedEffectiveTiming": "END_OF_CURRENT_CYCLE",
    "scheduledCommitAt": "2026-05-31T23:59:59+03:00",
    "scheduledAt": "2026-05-20T14:00:01+03:00",
    "unbindObsoleteEquipment": true,
    "obsoleteEquipmentList": [],
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER"
  }
}
```

Sample SubscriptionDowngraded (triple→double-play with STB return):

```
{
  "eventType": "SubscriptionDowngraded",
  "aggregateId": "sub_01HZ_OTIENO_FIBER",
  "timestamp": "2026-05-31T23:59:59+03:00",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_OTIENO_FIBER",
    "operationId": "subop_01HZ_OTIENO_DOWNGRADE_EOC",
    "downgradeTrigger": "CUSTOMER_REQUESTED_DOWNGRADE",
    "sourcePackageRef": "pkg_01HX_INET_VOIP_IPTV_RES",
    "sourcePackageVersionId": "pkv_01HX_INET_VOIP_IPTV_V1",
    "sourcePackagePrice": 7500.00,
    "targetPackageRef": "pkg_01HX_INET_VOIP_RES",
    "targetPackageVersionId": "pkv_01HX_INET_VOIP_V3",
    "targetPackagePrice": 4500.00,
    "currency": "KES",
    "cycleAnchorPolicy": "PRESERVE",
    "resolvedEffectiveTiming": "END_OF_CURRENT_CYCLE",
    "scheduledCommitAt": "2026-05-31T23:59:59+03:00",
    "creditDelta": 0.00,
    "creditBillableEventId": null,
    "oldCycleCreditAmount": null,
    "firstCycleCharge": null,
    "cycleStart": "2026-05-01",
    "cycleEnd": "2026-05-31",
    "billingFrequencyChanged": false,
    "droppedServiceClasses": ["IPTV"],
    "unboundEquipment": [{"equipmentId": "eqp_STB_01HZ_LAVI_042", "fulfilledServiceClass": "IPTV"}],
    "downgradedAt": "2026-05-31T23:59:59+03:00",
    "initiatingActorUserId": "cc-agent-wik-007",
    "initiatingActorRole": "CALL_CENTER"
  }
}
```

7. Cross-DD coordination

7.1 Upstream callers

Caller	downgradeTrigger	Role
Customer portal	CUSTOMER_REQUESTED_DOWNGRADE	CUSTOMER_SELF_SERVICE (subject to config)
Call Center	CUSTOMER_REQUESTED_DOWNGRADE	CALL_CENTER

Caller	downgradeTrigger	Role
BO admin	ADMIN_DOWNGRADE	SUBSCRIPTION_ADMIN or SUBSCRIPTION_OPERATOR

7.2 Sibling modules

Module	Direction	Contract
SUB-WF-FRAMEWORK-01 (parent)	This DD composes framework primitives	subscription_operation row + BPMN start
SUB-WF-FRAMEWORK-01-WORKERS	This DD's BPMNs reference workers by topic	Workers used: drools-decide, sub-lm-put-pending-status, sub-lm-compute-cycle-window, bil01-emit-intent, ful-call-modify, osr-equipment-unbind, sub-lm-put-master-fields, sub-lm-commit-state, kafka-emit, notification-send, subscription-operation-finalize
SUB-LM-01	Workers write status + master fields	Status flip + package_ref/version + previous_package_ref/version
SIP-01	Workers read Package + Package Version catalog	Target Package validation
PLM-CFG-01	Workers read service_class definitions	Equipment-drop delta computation
RLM-CFG-01	Workers read HomePass service_management_endpoints + status_code	HomePass physical-capability check; read-only
BIL-01	bil01-emit-intent worker	DOWNGRADE_INTENT with credit math
BIL-CFG-01	BillableEvent catalog	Downgrade-credit definitions
FUL-03	ful-call-modify worker	MODIFY_SERVICE. Owns per-service-class fan-out via HomePass service_management_endpoints
OSR-INSTANCE-01	osr-equipment-unbind worker	Equipment unbind (mark as returned-to-inventory)
OSR-RMA-01	Consumes SubscriptionDowngraded event with unboundEquipment list	Kicks off equipment-return WO (RMA flow)
NOT-01	notification-send worker	Customer notification per operator config
BIL-02 / CVM / EM-04	Consume Kafka events	New-price cycle accrual, churn/retention analytics, reporting

7.3 Build order

This DD deploys after:

1. SUB-WF-FRAMEWORK-01 (parent + WORKERS) v1.0
2. SUB-LM-01 v1.0
3. SIP-01 v1.0
4. PLM-CFG-01 v1.0
5. RLM-CFG-01 v1.0
6. FUL-03 v1.0
7. BIL-01 (DOWNGRADE_INTENT support)
8. OSR-INSTANCE-01 v1.0
9. OSR-RMA-01 v1.0 (consumer of equipment-return events)
10. At least one FLOW satellite (FLOW-WIK v1.0)

7.4 Stub debt + open coordination items

- **WUG / WTZ / YASSN FLOW satellites** — pending operator deep dives.
- **BIL-CFG-01 DOWNGRADE_INTENT BillableEvent definitions** — pending BIL-CFG-01 v1.0 promotion. Credit formula owned by BIL-01.
- **FUL-03 MODIFY_SERVICE intent + per-service-class fan-out** — pending FUL-03 v1.1 promotion (currently v1.0 ships APPLY_RESTRICTION/LIFT_RESTRICTION/SUSPEND_SERVICE under FUL-04; UPGRADE + DOWNGRADE need

MODIFY_SERVICE).

- **OSR-RMA-01 equipment-return WO integration** — pending OSR-RMA-01 consumer subscription to SubscriptionDowngraded Kafka event for kicking off pickup WOs.
- **PLM-CFG-01 service_class.requires_equipment field** — pending PLM-CFG-01 v1.0 promotion.

B — Current TZ

Status: To be filled in after codebase cartography review.

Legacy package downgrade flow exists; specific implementation shape — proration credit calculation, equipment-return propagation, network teardown sequencing, IPTV multicast group leave handling — requires source-code review to document accurately. Until that cartography lands, this DD does not assert specific legacy structural claims.